AN-107 I²C Bus Design Notes

Halcyon Micro — fictional company, synthetic document

Device addresses

Avoid address conflicts by consulting the table below before connecting devices to the same bus.

Device	7-bit address(es)	Selection
HM5V-PD	0x28 or 0x29	ADDR pin: GND = 0x28, VDD = 0x29
HT410	0x48, 0x49, 0x4A, 0x4B	ADD0 pin: GND, VDD, SDA, SCL
HIMU6	0x6A or 0x6B	SDO pin: GND = 0x6A, VDDIO = 0x6B
HAC24	0x1A or 0x1B	CS pin: GND = 0x1A, DVDD = 0x1B
HB7	—	Acts as I²C master only

Pull-up resistor sizing

The maximum pull-up value is RP(max) = tr / (0.8473 × CB), where tr is the allowed rise time (300 ns for 400 kHz Fast-mode, 120 ns for 1 MHz Fast-mode Plus) and CB is the bus capacitance. For CB = 100 pF at 400 kHz, RP(max) is about 3.5 kΩ; 2.2 kΩ is a good default at 3.3 V.

Speed limits

HT410, HM5V-PD and HB7 support up to 400 kHz. HIMU6 supports up to 1 MHz. Do not mix 1 MHz devices on a bus with 400 kHz devices unless the master runs at 400 kHz.